

# ECE 447

## Lesson 08

### Amplitude

### Modulation, Part 1



UNITED STATES  
AIR FORCE  
ACADEMY

# SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
  - **HW 1 due tonight.** Any issues?
  - **Lab 2 Assignment.** The assignment associated with Lab 2 is due Lesson 10 (Tue, 1 Sep) by 2359 via Gradescope upload.
  - **HW 2 assigned next lesson.** Only two problems - shorter HWs, assigned more often. Don't get behind! HWs and Labs reinforce the lesson material.

# DID YOU DO THE READING?

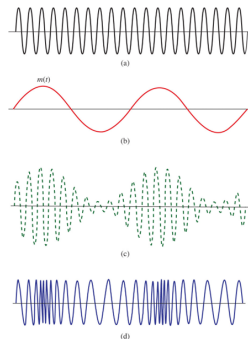
Where do we see *baseband communications*?

Where do we see *carrier communications*?

*Modulation* means *changing* something about a signal - and not always just shifting the signal's frequencies

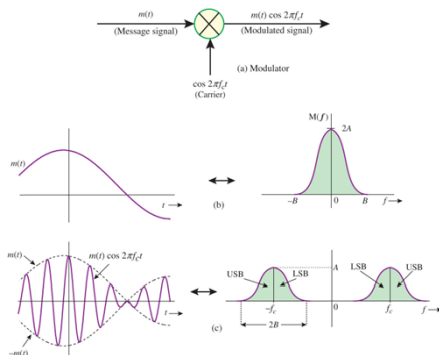
# MODULATION

- Message signal:  $m(t)$
- General carrier modulated signal:  
 $s(t) = A(t) \cos [2\pi f_c t + \phi(t)]$
- Amplitude (linear) modulation,  $A(t)$ 
  - $s_1(t) = m(t) \cdot \cos (2\pi f_c t)$
  - $S_1(f) = \frac{1}{2}M(f - f_c) + \frac{1}{2}M(f + f_c)$
- Angular modulation,  $f_c$  or  $\phi(t)$ 
  - Message modulates frequency or phase
  - Notice the general equation doesn't have  $f_c(t)$  - FM is a little different - more to come in the angle modulation lessons!



# DOUBLE-SIDEBAND, SUPPRESSED-CARRIER (DSB-SC)

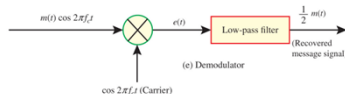
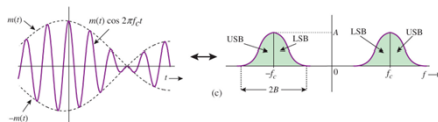
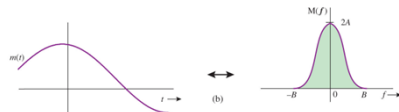
- Simplest form of AM
- Learned this in ECE 215
- $\varphi_{DSB-SC}(t) = A_c m(t) \cos(\omega_c t)$
- Over-, under-, or 100%-modulated?



# DOUBLE-SIDEBAND, SUPPRESSED-CARRIER (DSB-SC)

## • Demodulation

- Same as modulation with LPF
- What if  $f_c < B$ ?
- Reality,  $f_c/B \gg 1$  to avoid distortion
- Skills Review, Prob 6 - what happened when the receiver didn't have the exact frequency/phase?
- Difficult to achieve synchronized carrier signals for receiver



# EXTRA TIME?

Free EI!